UNIVERSITY OF RWANDA

RWANDA NZIZA

RWANDA NZIZA CITIZEN REPORTING SYSTEM

PROJECT TECHNICAL REPORT

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Report Contents

Chapter 1: Introduction

1.1 Historical Background

Rwanda promotes clean communities, accountable public services and digital communication. Citizens often observe damaged roads, water leaks, waste, electricity faults and other public problems. Rwanda Nziza was created as a simple website for reporting these issues and finding the responsible institution.

1.2 Problem Statement

Public problems may be reported with incomplete details or to the wrong institution. This can delay action and make follow-up difficult. The project provides one reporting form, an institution directory and a personal report page.

1.3 Objectives

The main objective is to build a simple citizen reporting website using basic HTML, CSS and JavaScript.

- Allow users to sign up and log in with phone number or email.
- Provide a searchable directory of at least fifty institutions.
- Allow users to submit a problem with location, description and image input.
- Allow users to view their reports and current status.
- Maintain simple design and navigation on all pages.

1.4 Expected Outcome and Importance

The result is a working educational prototype that demonstrates forms, navigation, search, events and browser storage. It can help citizens organize reports and identify the correct public authority.

Chapter 2: System Analysis and Design

2.1 System Analysis

2.1.1 Intended Users

- Citizens reporting community problems.
- Registered users checking their reports.
- Students and lecturers evaluating the project.
- Public institutions in a future connected version.

2.1.2 Functional Requirements

Function	Purpose
Sign up and login	Create an account and access personal reports.
Institution search	Find institutions using typed words.
Problem report	Save contact, category, location, image name and description.
My Reports	Display report ID, date, institution, location and status.
Navigation	Connect all pages and support logout.

2.2 System Design

HTML provides content, CSS controls appearance and JavaScript handles interaction. User and report data are stored in browser localStorage.

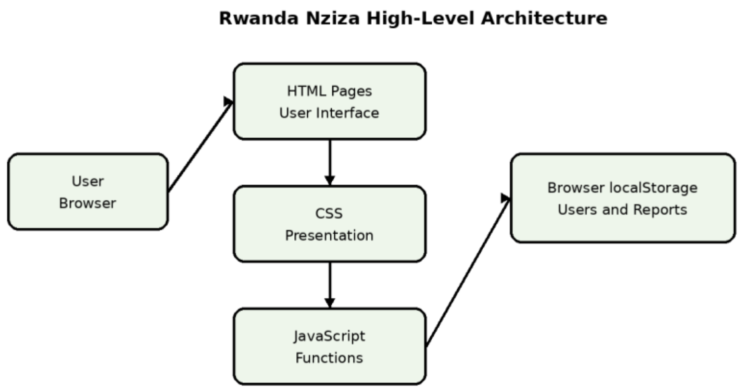

Figure 1: High-level architecture.

2.2.1 Main User Flow

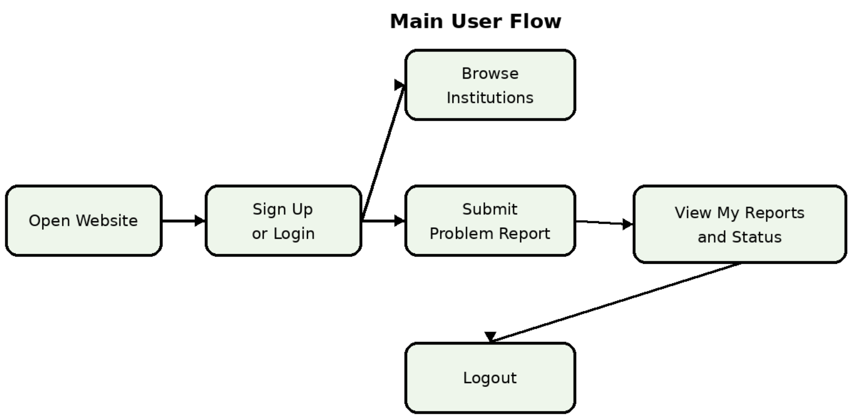

Figure 2: User navigation flow.

2.2.2 Data Design

Record	Main fields	Relationship
User	Full name, phone, email, password	One user can submit many reports.
Report	ID, user, category, institution, location, image, description, date, status	Each report belongs to one user.

Chapter 3: Implementation

3.1 Technologies

The project uses basic HTML, CSS and JavaScript. Images are stored in the Media folder, and all pages share one stylesheet and one script file.

3.2 Main Pages

Page	Main purpose
Home	Introduces Rwanda Nziza and links to reporting and institutions.
Institutions	Shows at least fifty searchable institutions and districts.
Report a Problem	Collects identity, category, institution, location, image and description.
Sign Up	Collects full name, phone number, email and password.
Login	Accepts email or phone number with password.
My Reports	Shows the logged-in user's reports and status.
About	Explains the vision, mission and contact information.

3.3 Main JavaScript Functions

- Open and close the mobile menu.
- Search institution cards.
- Count report-description characters.
- Register users and validate login.
- Save reports and display them for the current user.
- Update report counter, date, time and logout links.

3.4 Current Limitation

The system is a front-end prototype. Data is saved only in the browser, so it is not shared between devices and is not yet sent to real institutions.

Chapter 4: Conclusion and Recommendations

4.1 Conclusion

Rwanda Nziza meets the main project objectives. It provides connected pages, account access, institution search, report submission, image input and personal report tracking. Most prototype features are complete.

4.2 Recommendations

- Add a secure server and relational database.
- Encrypt passwords and validate data on the server.
- Allow institutions to update report status.
- Add email or SMS notifications and map location.
- Train users to submit accurate information and clear images.

Appendix

Project repository or deployment link: https://github.com/fideleigeno/RwandaNziza

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